

## Lab. 1 – Numerical analysis of TLs

**Assigned:** Friday March 20, 2026  
**Due:** Friday April 3, 2026

### Notes

To take the exam with the grading of submitted assignments + oral discussion it is necessary:

- to prepare **individually** the report of the lab, that must include the Matlab script, the obtained results and their brief comment;
- to submit the report **within the deadline**, uploading the PDF in the section “Elaborati” of the web portal

Let assume that the transmission lines designed/analyzed in problem 1 and problem 2 of the Assignment 1 are realized with microstrip technology.

The substrate is an FR4 material, with relative dielectric constant  $\epsilon_r = 4.45$  and thickness  $h = 1.55\text{mm}$ . The central frequency is  $f_0 = 5.8\text{ GHz}$ .

Repeat the analysis carried on in problem 1 and problem 2 of the assignment 1 with Matlab™ of the assignment 1 using CST.

In particular:

- calculate the length and width of the different lines (for the given values of the characteristic impedances) and implement the geometry in CST;
- run the solver;
- compare the results with those obtained with Matlab™ (export the data from CST, import them in Matlab™, and represent the two sets on the same plot).